

Background Information

This course is intended to enhance the learning experience of the student in topics encountered in Eeng-1082. In this lab, students are expected to get hands-on experience in using the basic measuring devices used in electrical engineering and in interpreting the results of measurement operations in terms of the concepts introduced in the first electrical circuits course. How the student performs in the lab depends on his/her preparation, participation, and teamwork. Each team member must participate in all aspects of the lab to insure a thorough understanding of the equipment and concepts. The student, lab teaching assistant, lab instructor and faculty coordinator all have certain responsibilities toward successful completion of the lab's goals and objectives.

Student Responsibilities

The student is expected to be prepared for each lab. Lab preparation includes reading the lab experiment and related textbook material. Active participation by each student in lab activities is expected. The student is expected to ask the teaching assistant or lab instructor any questions he/she may have. Do Not Make Costly Mistakes Because You Did Not Ask A Simple Question. Moreover, understanding the concepts and procedure of each lab is necessary for successful completion of the lab.

The student should remain alert and use common sense while performing a lab experiment. He/she is also responsible for keeping a professional and accurate record of the lab experiments in a laboratory notebook. Students should report any errors in the lab manual to the teaching assistant or lab instructor.

As students are expected to abide by the code of conduct of students (article 166 and 166.1.1 of the senate legislation of Bahir Dar University May 20, 2005) throughout this course

- Academic dishonest, including cheating, fabrication, and plagiarism will not be tolerated.
- 100 % of class attendance is mandatory!
- Cell phone MUST be turned off before entering the lab class
- Chewing gum will not be tolerated

Laboratory Teaching Assistant and Lab Instructor Responsibilities

Both the Laboratory Teaching Assistant (LTA) and lab instructor shall be completely familiar with each lab prior to class. The LTA shall provide the students with a safety review during the first class. The LTA is responsible for insuring that all the necessary equipment and/or preparations for the lab are available and in working condition. LAB EXPERIMENTS SHOULD BE CHECKED IN ADVANCE TO MAKE SURE EVERYTHING IS IN WORKING ORDER. The lab instructor should fully answer any questions posed by the students and supervise the students performing the lab experiments. The lab instructor is expected to grade the lab notebooks and reports in a fair and timely manner. The reports should be returned to the students in the next lab period following submission. The LTA and lab instructor should report any errors in the lab manual to the faculty coordinator.

Faculty Coordinator Responsibilities

The faculty coordinator should insure that the laboratory is properly equipped, i.e., that the teaching assistants receive any equipment necessary to perform the experiments. The coordinator is responsible for supervising the teaching assistants, lab instructors and resolving any questions or problems that are identified by the teaching assistants and lab instructors or the students. He/she is also responsible for making any necessary corrections to this manual. The faculty coordinator is responsible for insuring that the software version of the manual is continually updated and available.

1. Requirements

i. Attendance

A laboratory is a practical experience requiring proper equipment and involving team work. Therefore attendance at all laboratory sessions is mandatory. Students who miss a laboratory session must take up for it at the first opportunity and arrange it with the instructor. If the instructor is more than 15 minutes late, students may leave the lab.

ii. Deliverables

On time submission of Lab report should be mandatory and as per the requirement of the instructor but commonly, reports are completed at home by each student individually or in group (as per requirement) and are handed to the instructor prior to the next Lab work.

2. Safety Measures

Electricity, when improperly used, is very dangerous to people and to equipment. This is especially true in Electrical laboratories and an industrial environment where large amounts of power are used. In an experimental or teaching laboratory here inexperienced personnel may use electrical equipment in experimental or nonstandard configuration.

The knowledge and habit-forming experience to work safely around electrical equipment and the ability to design safe electrical equipment begins with the first student laboratory experience and continues through life. This includes learning the types of electrical injuries and damage, how they can be prevented, the physiology of electrical injuries, and steps to take when accidents occur.

Physiology of Electrical Injuries:

There are three main types of electrical injuries: electrical shock, electrical burns, and falls caused by electrical shock. A fourth type, 'sunburned' eyes from looking at electric arcs, such as arc-welding, is very painful and may cause loss of work time but is usually of a temporary nature. Other injuries may be indirectly caused by electrical accidents. Under some circumstances death can occur from voltages from substantially less than the nominal 120 Volts.

Electric shock is not a single phenomenon but is a disturbance of the nerves that is caused by electric current. If a victim receives an electric shock from grasping a live conductor, a current of greater than 15 to 30mA through the arm will cause muscle contractions so severe that the victim cannot let go.

Source of Electric Shock:

Since electric shock is caused by an electric current through a part of the body, it is prevented by not allowing the body to become part of any electric circuit. From this viewpoint, electric circuits may be classified as either grounded or ungrounded.

Residential, commercial, and industrial systems such as lighting and heating are always grounded for greater safety. Ungrounded circuits are used in systems where isolation from other systems is necessary, where low voltages and low power are used, and in other instances where obtaining a ground connection is difficult or impractical. Thus, touching a supposedly safe conductor while standing on the ground would result in an electric shock.

Prevention of Shock in the Laboratory:

Prevention of electric shock to individuals and damage to equipment in the laboratory can be done by strict adherence to several common-sense rules summarized below:

1. When hooking up a circuit, connect to the power source last, while power is off.
2. Before making changes in a circuit, turn off or disconnect the power first, if possible.
3. Never work alone where the potential of electric shock exists.
4. When changing an energized connection, use only one hand. Never touch two points in the circuit that are at different potentials.
5. Know that the circuit and connections are correct before applying power to the circuit.
6. Avoid touching capacitors that may have a residual charge. The stored energy can cause a severe shock even after a long period of time.
7. Insulate yourself from ground by standing on an insulating mat where available.

After Accident Action:

Since accidents do happen despite all efforts to prevent them, plans for appropriate reaction to an accident can save time and lives. Such a plan should include immediate availability of first aid material suitable for minor injuries or for injuries that are likely because of the nature of the work. Knowledge of how to obtain trained assistance such as Emergency Medical Services (EMS) should be readily available for everyone.

Treating victims for electrical shock includes four basic steps that should be taken immediately. Step two requires qualification in CPR and step three requires knowledge of mouth-to-mouth resuscitation. Everyone who works around voltages that can cause dangerous electrical shock should take advantage of the many opportunities available to become qualified in CPR and artificial respiration.

Immediate Steps after Electric Shock:

1. Shut off all power and remove victim from the electric circuit. If the power cannot be shut off immediately, **use an insulator** of some sort, such as a wooden pole, to remove victim from the circuit. **Attempts to pull the victim from the circuit with your hands will almost always result in your joining the victim in the electric shock.**
2. If you are qualified in CPR, check for ventricular fibrillation or cardiac arrest. If either is detected, external cardiac massage should be started at once. Whether you are qualified in CPR or not, notify EMS and the ECE Department at once, using the telephone numbers listed below.
3. Check for respiratory failure and take appropriate action. This may have resulted from physical paralysis of respiratory muscles or from a head injury. Sometimes many hours pass before normal respiration returns. **Artificial respiration should be continued until trained EMS assistance arrives.**
4. Check for and treat other injuries such as fractures from a fall or burns from current entry and exit sites. Investigations are always after accidents. As an engineer you will be involved as a part of the investigating team or in providing information to an investigator. Information obtained and notes written immediately after the emergency will aid this investigation and assist in preventing future accidents of a similar nature.

Instrument Protection Rules:

1. Set instrument scales to the highest range before applying power.
2. Be sure instrument grounds are connected properly. Avoid accidental grounding of "hot" leads, i.e., those that are above ground potential.
3. Check polarity markings and connections of instruments carefully before connecting power.
4. **Never connect an ammeter across a voltage source.** Only connect ammeters in series with loads.
5. Do not exceed the voltage and current ratings of instruments or other circuit elements. This particularly applies to Wattmeter's since the current or voltage rating may be exceeded with the needle still on the scale.
6. Be sure the fuse and circuit breakers are of suitable value. When connecting electrical elements to make up a network in the laboratory, it is easy to lose track of various points in the network and accidentally connect a wire to the wrong place. A procedure to follow that helps to avoid this is to connect the main series part of the network first, then go back and add the elements in parallel. As an element is added, place a small check by it on your circuit diagram. Then go back and verify all connections before turning on the power. *One day someone's life may depend upon your making sure that all has been done correctly.*

4. The Laboratory Report

The report you submit after completion of each series of experiments is the main product of the team's work which will be used for grading. Therefore you should devote enough attention to this final but critical step in the laboratory experience.

The purpose of the laboratory report is to provide information on the measurement procedure, obtained results, analysis, and interpretation and discussion of these results. The discussion and conclusions are very important in a report because they show what knowledge you gained by doing the experiments.

There is no one best format for all technical reports but there are a few simple rules concerning technical presentations which should be followed. Adapted to this laboratory they may be summarized in the following recommended report format:

1. Cover page
2. Introduction
3. Experimental Procedure
4. Experimental Data/ Observation
5. Discussion and Analysis
6. Conclusions and Recommendation

Detailed descriptions of these items are given below.

1. **Cover Page** should have the names of the University/ Institute, School/Faculty, team members, and a designation such as Group1, 2 or 3. Group members should be listed in alphabetical order and should be numbered; also it should include section and course numbers (e.g. Eeng1082, Section 001). Moreover, it should contain laboratory number and title of the experiment, such as "Lab No X", "Ohms Law". Cover page should also contain to whom to submit and when the report is delivery.
2. **Introduction** should contain a brief statement or theory in which you state the objectives, or goals of the experiments. It should also help guide the reader through the report by stating, for example, that experiments were done with three different circuits or consisted of two parts etc. It is also a good place to state that the experimental results were as expected or that there were

some problems, as explained in the conclusions. You may state what was learned in this set of experiments

3. **Procedure** describes the experimental setup and how the measurements were made. Include here circuit schematics with the values of components. Mention instruments used and their settings. Procedures should be written in passive voice form.
4. **Experimental Data** section should be presented clearly with a reference to the procedure the schematic used in measurements. Tables are often a good way of presenting results. This section can also include some calculations or data analysis.
5. **Discussion and Analysis:** it is a critical part of the report which testifies to the student's understanding of the experiments and its purpose. In this part of the report you should compare the expected outcome of the experiment, such as derived from theory or computer simulation, with the measured value.
6. **Questions:** Questions pertaining to the lab should be answered here.
7. **Conclusions** should contain several short statements closing a report. They should inform the reader if the experiments agreed with the theory. If there were differences between measured and expected results, try to explain possible reasons for these differences. You may also say what could have been done differently, how experiments may be improved, or make other comments on the laboratory.

6. Other Resources

The manual itself does not contain all information needed in the laboratory. You are expected to consult the literature or your own lecture notes, as needed. Any good book on Electrical Power System will do.

- Highly recommended: Introductory Circuit Analysis by Boylestad, 10th Edition.

Laboratory 1

Equipment Familiarization

1. Introduction

In the first lab period, the students should become familiar with the location of equipment and components in the lab, the course requirements, and the teaching instructor. Students should also make sure that they have all of the co-requisites and pre-requisites for the course at this time.

2. Objective

To familiarize students with the lab facilities, equipments, standard operating procedures, lab safety, and the course requirements

3. Equipment Required

- Eeng - 1082 lab manual
- Function Generator
- Digital Oscilloscope
- DC Power Supply
- Digital Multimeter
- Breadboard
- Resistors
- Capacitors
- Inductors

4. Procedures

1. During the first laboratory period, the instructor will provide the students with a general idea of what is expected from them in this course. Each student may receive a copy of the syllabus. In addition, the instructor will review the safety concepts of the course.
2. The instructor will indicate which word processor should be used for the lab reports. The students should familiarize themselves with the preferred word processor software.

3. During this period, the instructor will briefly review the equipment which will be used throughout the semester. The location of instruments, equipment, and components (e.g. resistors, capacitors, connecting wiring) will be indicated. The guidelines for instrument use will be reviewed.

Laboratory 2

Resistors and the Resistor Color Code

1. Background Theory

An important lesson to learn in electronics is how to read the color codes for resistors. If you plan a career in electronics, it will help you to memorize the color code, and the technique for calculating the resistor value. This laboratory will teach you how to read resistor color codes.

Reading the resistor color code is easy with a little practice. Most resistors have four color bands, while some have five bands. The chart below (Figure 1) shows the color codes and their respective values depending on which band they fall on. The resistor value is decoded by reading the colors from left to right. The tricky part is determining which is the left side is and which the right side is. Do this by finding the gold or silver band which is always on the right side. Then start reading the resistor colors from the left.

- ✓ Look at a common 1K resistor. From left to right, a 1K resistor will have **brown-black-red-gold**. Looking at Figure 1, this decodes respectively to 1 - 0 - $\times 100$ - $\pm 5\%$.

Take the first and second significant digits together to be 10. Then multiply by the multiplier 100. That gives you 1,000 ohms as the resistor value, which is 1K. So what's with the ± 5 ? The tolerance band tells us that the measured resistance can be off by plus or minus 5%. So the actual measured resistor value could be anywhere from 950 ohms to 1050 ohms.

- ✓ Now look at a 27 K resistor. Again from left to right it reads red-violet-orange-gold which translates to 2-7 - $\times 1000$ - ± 5 . So this would be $27 \times 1000 = 27,000$ ohms which is 27 K.

The fifth band is rarely used on resistors, but is included in the color code chart so that you have a more complete reference. The fifth color band signifies the rate at which the resistor fails when operating at its full rated wattage. It is read as a percentage failure per 1,000 hours.

For practice, try reading color codes from resistors with unknown values. Try to figure out what the resistor value should be from the color code. Then measure the resistor with an ohmmeter or multimeter to see if you got it right. With some practice, you will be able to pick out common resistors from a box of spare parts just by glancing at the color bands.

Color	Digits	Multiplier	Tolerance	Thermal Coefficient
Black	0	1		
Brown	1	10	1%	100ppm
Red	2	100	2%	50ppm
	3	1k		15ppm
Yellow	4	10k		25ppm
	5	100k	0.5%	
Blue	6	1M	0.25%	
Violet	7	10M		
Gray	8			
White	9			
Gold			5%	
Silver			10%	

Figure 1: Resistor color coding

To understand the numbering skins, look the example shown below:

		
6-band color code	5-band color code	4-band color code
3 digits, multiplier, tolerance, thermal coefficient	3 digits, multiplier, tolerance	2 digits, multiplier, tolerance

Figure 2: Resistor color coding number skins

Finally you can see the normal designation as shown in the table below:

Band designation for the Resistor Color Codes					
Color	First Band	Second Band	Third Band	Fourth Band	Fifth Band
	First Significant Figure	Second Significant Figure	Multiplier	Tolerance (%)	Fail Rate (% per 1000 hrs)
Black	-	0	$\times 1 = 10^0$		
Brown	1	1	$\times 10 = 10^1$		1.0
Red	2	2	$\times 100 = 10^2$		0.1
Orange	3	3	$\times 1000 = 10^3$		0.01
Yellow	4	4	$\times 10000 = 10^4$		0.001
Green	5	5	$\times 100000 = 10^5$		
Blue	6	6	$\times 1000000 = 10^6$		
Violet	7	7	$\times 10000000 = 10^7$		
Gray	8	8			
White	9	9			
Gold	-	-	-	± 5	
Silver	-	-	-	± 10	
No Color	-	-	-	± 20	

2. Objectives

- To study various types of resistors used in electrical circuit and codes.
- To study ohmmeter and their use in determining resistor.
- To determine the ohmic value, tolerance, and reliability of carbon compositions resistors using color code
- To study relation between wattage and resistance of a resistor and determine their resistance value.

Resistance

3. Materials Required

- ✓ Ohmmeters
- ✓ Different color-coded resistors
- ✓ Circuit construction board
- ✓ Probes (connecting wires) if it is needed.

4. Procedures

1. Arbitrarily assign numbers for each resistor
2. Comfortably mount each resistor on the circuit construction board
3. Record the color of each band for resistor # 1 in the appropriate horizontal columns in Data table 1-1 then indicate the resistors color-coded ohmic value in vertical column labeled Res (ohms)
4. Calculate the resistors upper and lower tolerance limit and record in the appropriate vertical columns. Show a sample calculation.
5. Measure the ohmic value of # 1 resistor using ac ohmmeters properly and record in vertical column labeled Resistor (ohm).
6. When applicable record the resistors reliability in the last vertical column.
7. Repeat steps 2-6 for each of the remaining resistors.

Data Table 1-1:

Data		Color Code				Calculated				Meas.
No	1st	2nd	3rd	4th	5th	Res (Ω)	Tol. (%)	Upper limit	Lower limit	Res (Ω)
1										
2										
3										
4										
5										
6										
7										
8										
9										
10										

5. Review Questions

1. Predict the color coding of 870Ω , 9.2Ω and 0.50Ω resistors.
2. Differentiate the reliability and tolerance of a carbon composition resistor?
3. Compare the Res (ohm) and the Meas. (ohm) column from the Data-table and come to a conclusion.
4. How does the physical size of a resistor relate to its wattage rating?
5. Discuss the effect of temperature on the resistance.

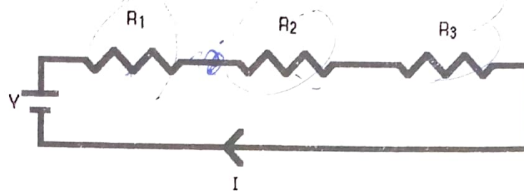
Laboratory 3

Resistors in Series, In Parallel and Both Combinations

1. Background Theory

• Series Circuits

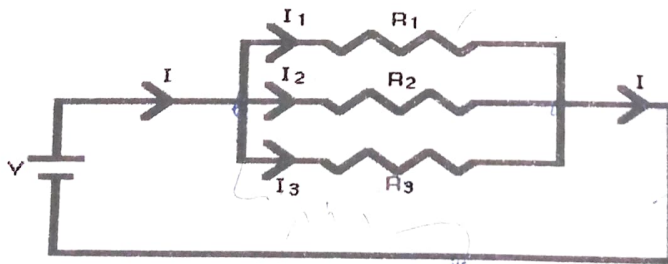
A series circuit is a circuit in which resistors are arranged in a chain, so the current has only one path to take. The current is the same through each resistor. The total resistance of the circuit is found by simply adding up the resistance values of the individual resistors: Equivalent resistance of resistors in series: $R = R_1 + R_2 + R_3 + \dots$



• Parallel circuits

A parallel circuit is a circuit in which the resistors are arranged with their heads connected together, and their tails connected together. The current in a parallel circuit breaks up, with some flowing along each parallel branch and re-combining when the branches meet again. The voltage across each resistor in parallel is the same. The total resistance of a set of resistors in parallel is found by adding up the reciprocals of the resistance values, and then taking the reciprocal of the total:

Equivalent resistance of resistors in parallel: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$



- Most electronic circuits consist of combination of series and parallel circuits called series-parallel circuits. When dealing with these circuits it is necessary to reduce each combination to

an equivalent resistance. These are then added to the series elements in the circuit to determine the total resistance.

2. Objectives

- To study series, parallel and series -parallel circuit connection of the resistors in an electrical circuit.
- Determination of equivalent resistance of a series, parallel and series parallel circuit and to verify the result by theoretical calculation.

3. Materials Required

- Resistors with different value
- Circuit construction board
- Digital Multi-meter
- Connecting wires

4. Procedure (a)

1. Construct the series circuit shown in *figure a* and *figure b*.



Figure a



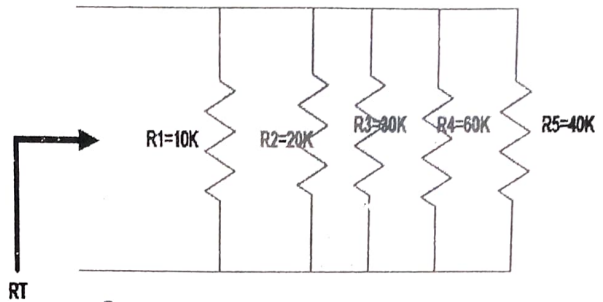
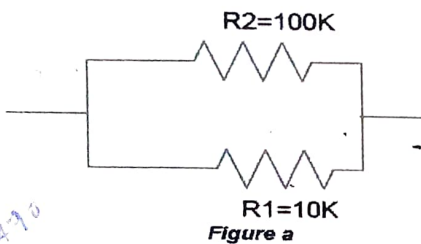
Figure b

2. Calculate the circuit's total resistance (R_T) and record in a proper table.
3. Measure the circuit's resistance and record in the same table.
4. Calculate the percentage difference b/n calculated and measured values of R_T and record in the same table. Show your calculation.

5. Remove the components for the circuit in figure a. from your board and construct the series circuit shown in figure b.
6. Repeat steps a-c above.

5. Procedure (b)

1. Construct the parallel circuit shown in figure a.



2. Calculate the circuit's total resistance (R_T) and record in the proper table. Show your calculation.
3. Measure the circuit's resistance and record in the same table.
4. Calculate the percentage difference between calculated and measured values or R_T and record in the same table. Show your calculation.

$$\% \text{ difference} = \frac{\text{calculated } R_T - \text{Measured } R_T}{\text{Calculated } R_T}$$

5. Remove circuit Figure a from your board and construct the parallel circuit shown in Figure b.
6. Repeat steps 2-4 above.

6. Procedure (c)

1. Construct the series parallel circuit as shown below figure a.

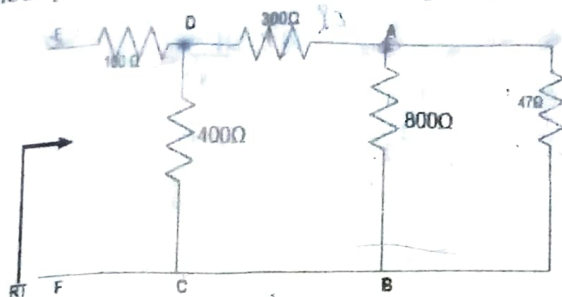


Figure a

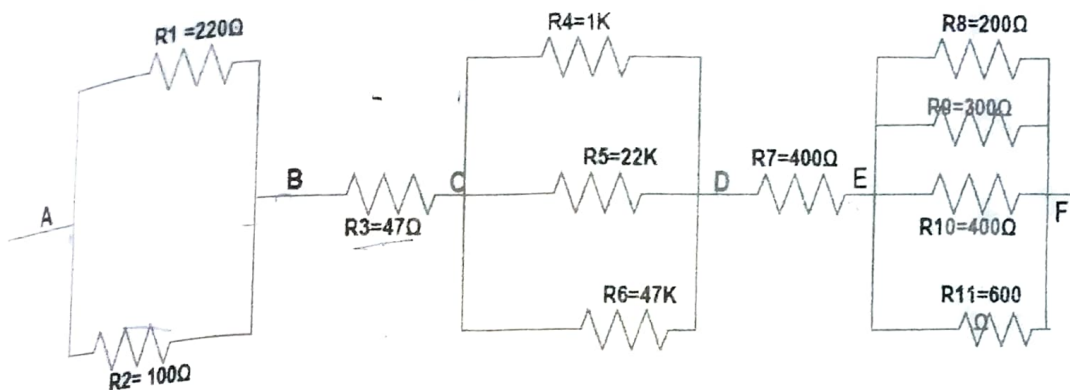


Figure b.

2. Calculate the equivalent resistance at each junction in figure a. and record the result in the appropriate columns in data table 2.1. Calculate also the total resistance and record in the same data table. Show your calculations.
3. Measure the equivalent resistance at each junction and the total circuit resistance for figure a and record in the appropriate columns in data table 2.1.
4. Remove circuit figure a from your board and construct the circuit shown in figure b.
5. Calculate the parallel resistances (R_{a-b} , R_{c-d} and R_{e-f}) and the total circuit resistance and record this data in the appropriate column in data table 2.2. Show your calculation.
6. Measure the parallel resistance (R_{a-b} , R_{c-d} and R_{e-f}) & the total circuit resistance and record in the appropriate column in data table 2.2.
7. Calculate the difference between the measured and calculated values.

9.10.10
E=

series
resistor
combination of
series & parallel

Table 2.1

	Circuit Junction	R Calculated	R measured	%diff
Series Branch	A-B			
	C-D			
	E-F			
Parallel Branch	A-B			
	C-D			
	E-F			

Table 2.2

R_{A-B}	Calc: Meas:
R_{C-D}	Calc: Meas:
R_{E-F}	Calc: Meas:
R_T	Calc: Meas:

7. Review Questions

1. What do you understand from the above experiments?
2. If you want to get the smallest resultant resistance from two or more resistances, which type of connection, do you choose?
3. If you want to get the highest resultant resistance from two or more resistances, which type of connection, do you choose?
4. Describe the advantage and disadvantage of series connection.
5. Describe the advantage and disadvantage of parallel connection.
6. Is the real world load purely series, parallel or series-parallel type.

Laboratory 4

Ohm's Law

1. Background Theory

The rate of flow of electricity in a given circuit is called current, denoted as I , the potential difference between the starting and the ending points of the circuit is known as the voltage denoted as V and the opposition to the flow of current is called resistance, denoted as R . According to **Ohms law** the ratio, V and I is always a constant factor for a particular conductor when the temperature, length, and conductor material is kept constant. This constant is called the Resistance that is characteristic of the conductor used. This is denoted by the formula

$$\frac{\text{Voltage}}{\text{Current}} = \text{constant} ; \quad \frac{V}{I} = R$$

The unit of EMF is Volts, the unit of Current is Amperes and the unit of Resistance is Ohms. Electricity is measured in these units. Ohms law has wide applications in electrical circuits obeying Kirchhoffs Laws, heat generation, Chemical analysis deposits, and most importantly for deriving Light energy.

Ohms Law for DC circuits is $R = \frac{V}{I}$ since the current flow is a steady one, but for AC, the current flow is a fluctuating one with some frequency, in such case the frequency is also taken into consideration, here the Resistance of the circuit is called the Impedance represented by Z . Ohms Law applies to the conductors whose resistance is independent of voltage. When an I, V graph is drawn, it is seen the Ohms law is obeyed in the linear portion for ohmic resistors, for non-ohmic resistance substances the I, V graph is bent and curved, showing negative resistance properties like incandescent lamps where more voltage is applied, more heat is generated and the resistance rises.

2. Objectives

- To study ohm's law and prove experimentally that current is proportional to the voltage across a dc circuit and to show that the proportionality constant is equal to the resistance of the circuit.
- To study relation between current and resistance in a dc and AC circuits for various voltages across the circuit.

3. Materials and Equipments Required

- Digital Voltmeters DC and AC
- DC Power supply
- Resistors variable in size
- Circuit construction board
- Connecting wires

4. Procedures

- Adjust the power supply to deliver 1V (or as close as possible) use separate dc voltmeters for this adjustment.

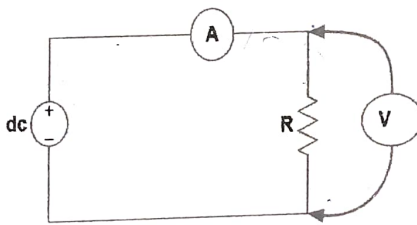


Figure a

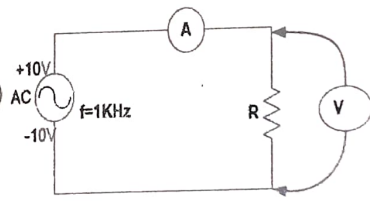


Figure b

- Construct the dc circuit shown in diagram Figure a, and connect it to the 1V power supply.
- Measure the voltage drop across R_1 (VR1) and record in data table 4.1 in the appropriate column and record I_T from the ammeter in I_T (meas.) column.
- Calculate I_T and record in the appropriate column in data table 4.1. Show your calculation.
- Repeat step 3 and 4 by increasing the power supply in steps of one volt
- Substitute the values of R_1 in data table 4.2 and calculate the corresponding circuit currents and record this information in the appropriate column in data table 4.2

7. Read the corresponding circuit current (I_T measure) and record this reading in the last column in data table 4.2.
8. Repeat the steps 1-7 for the AC circuit shown in *figure b*.

Data Table 4.1:

Source voltage	V_{R_1}	I_T (calc.)	I_T (meas.)
1	0.2	0.2	0.004
2			
3			
4			
5			
6			
7			
8			

Data Table 4.2:

$R_1 \Omega$	I_T (calc.)	I_T (meas.)
220 Ω		
330 Ω		
470 Ω		
1k Ω		
1.2k Ω		
2.2k Ω		
4.7k Ω		

5. Review Questions

1. Draw the graph of I_T Vs V_{RT} (calc.) on a millimeter paper.
2. Draw the graph of I_T Vs V_{RT} (meas.) on a millimeter paper.
3. What do you understand from your graph?
4. Does the graph linear or non linear?
5. If the graph is linear, what is the corresponding parameter in I_T Vs V_{RT} .
6. Discuss the difference between your results when you use the *dc* and *AC* sources.
7. Conclude any other relevant points based on this experiment.